A global multi-layer phonetic network in the Biblical Pentateuch

Eliahu Glikshtern

Independent Researcher

eliahu.glikshtern@gmail.com

ABSTRACT

The Biblical Pentateuch, traditionally classified as a legal and narrative text, has been observed in prior literary and recent computational work to exhibit systematic rhythmic and phonetic patterning in its recited form across the corpus.

The present study investigates how such large-scale patterning is structurally realized, using a reproducible phonetic representation and statistical framework applied to the full Pentateuch and compared against a corpus of modern Hebrew literary prose.

The analysis identifies a multi-layer phonetic system composed of interacting levels of organization, including consonantal recurrence, boundary-sensitive structuring, and distributed rhyme-like correspondences. These patterns are consistently present across all five books and remain robust under randomized controls that preserve local phonetic distributions.

Rhyme-like correspondences exhibit non-uniform, distance-selective recurrence, concentrating at specific preferred intervals, while simultaneously maintaining broad distribution across the text with very limited extended low-density regions. Phonetic dependencies are further aligned with structural segmentation, indicating coordination between phonetic and compositional units.

In contrast, modern Hebrew prose displays only weak and locally constrained analogues of these effects, lacking both the distance-selective structure and the global continuity observed in the Pentateuch.

Taken together, these findings support the interpretation that phonetic organization in the Pentateuch forms a coherent large-scale system that is not readily accounted for by local clustering or generic properties of the language alone. The observed combination of structured recurrence and global continuity is consistent with a systematically organized compositional framework operating across the corpus.

Keywords

phonetic system; multi-layer structure; Biblical text; Hebrew language; sound patterning; rhyme distribution; textual segmentation; computational analysis; digital humanities

1. INTRODUCTION

The study of phonetic and poetic structure in texts has traditionally focused on local phenomena, such as rhyme, meter, and line-based organization [1–3]. These features are typically analyzed within relatively small textual units and are understood as surface-level devices associated with explicitly poetic genres. As a result, large-scale phonetic organization spanning extended textual corpora has received comparatively limited systematic attention [4, 5, 11].

Existing approaches largely focus on local or line-based phenomena and do not provide a framework for detecting coordinated phonetic structure across continuous large-scale corpora.

The Biblical Pentateuch—traditionally classified as a legal and narrative text—has long been noted in literary and biblical scholarship to exhibit systematic rhythmic and phonetic patterning in its recited form [6–8]. These observations suggest that phonetic regularities may not be confined to isolated passages, but may instead reflect broader organizational structure. However, the internal organization and structural basis of such large-scale patterning remain unclear.

Related work has examined phonetic and rhythmic organization in the Pentateuch through the lens of cantillation, demonstrating systematic musical and structural organization in the traditional recitation system [10].

To our knowledge, the present study is the first systematic multi-layer quantitative investigation of large-scale phonetic organization across the full Pentateuch.

The present study addresses this question by testing whether extended phonetic patterning can be understood as the product of a structured system operating across the full text. To this end, a reproducible phonetic representation of the Hebrew corpus is constructed, and a set of statistical analyses is applied to detect and quantify phonetic dependencies at multiple levels.

Rather than treating these phenomena as independent effects, the analysis is organized around three interacting domains: consonantal recurrence, sensitivity to structural segmentation, and syllabic rhyme-like correspondence. These domains are investigated as potential components of a unified system of phonetic organization.

A central question is whether the observed patterns can be reduced to local properties of the language—such as morphology or phonological constraints—or whether they reflect higher-level organization at the scale of the text [4, 5, 9]. To evaluate this, randomized control models are employed that preserve local phonetic distributions while disrupting larger-scale structure. In addition, the Pentateuch is compared to a corpus of modern Hebrew literary prose, selected to approximate a comparable narrative style.

The results indicate that phonetic patterning in the Pentateuch is structured, non-uniform, and extends across textual boundaries while remaining robust under local-preserving randomization. In particular, rhyme-like correspondences exhibit non-uniform, distance-selective distributions together with continuous coverage across the text, consistent with organized large-scale structuring. Similar observations have been noted in other parts of the Hebrew Bible, though the present study focuses on the Pentateuch as a primary test case due to its scale and internal consistency.

For comparison, the analysis includes a corpus of modern Hebrew literary prose. While related phonetic tendencies are also detectable in this material, they appear in weaker and less consistently organized form, with only occasional local coordination.

Taken together, these observations motivate the hypothesis that extended texts may exhibit non-random phonetic structuring arising from the interaction of multiple structural layers. In this view, phonetic organization is not limited to isolated features, but reflects a constrained phonetic environment in which different levels of structure jointly contribute to large-scale patterning.

In modern terms, aspects of this organization may be compared to distributed rhyme systems found in certain oral-poetic forms, in which phonetic recurrence operates across multiple scales rather than within fixed line structures.

The present analysis focuses on the phonetic stream itself, without incorporating prosodic timing, stress, or performance-based alignment, in order to examine structural organization independently of recited rhythm.

2. Methods

2.1 Corpus and study design

The primary corpus consists of the Biblical Pentateuch (Torah), comprising the five books Genesis, Exodus, Leviticus, Numbers, and Deuteronomy. The text was obtained programmatically via the Sefaria API (version: *Tanach with Ta'amei HaMikra*), ensuring a fully vocalized Hebrew source including niqqud and cantillation marks at the raw stage.

Cantillation marks were removed during preprocessing, while phrase boundaries were retained and encoded using a standardized boundary delimiter. The resulting text was transformed into a continuous phonetic stream preserving word and phrase boundary structure.

The Pentateuch is treated as the principal test case due to its scale, internal consistency, and long-standing observations of systematic phonetic and rhythmic patterning in recited performance.

For comparison, a corpus of modern Hebrew literary prose was constructed from *HaGamad Baal HaChotem* (**Dwarf Long-Nose**; Ben Yehuda Project), a fully vocalized narrative text with niqqud and without cantillation marks.

The text was processed using the identical preprocessing and phonetic encoding pipeline as the primary corpus, ensuring full comparability of representation and analysis.

All analyses are defined within a unified computational framework consisting of deterministic preprocessing, phonetic encoding, recurrence-based metrics, controlled null models, and comparative evaluation. Randomized controls preserve local phonetic distributions while disrupting long-range structure, enabling isolation of large-scale organizational effects.

The study is computational and observational; no human subjects or experimental intervention are involved.

2.2 Text preprocessing

The Hebrew source text is normalized prior to phonetic encoding. The preprocessing pipeline includes retention of consonantal letters and vowel marks (niqqud), removal of cantillation marks, normalization of punctuation and phrase-final markers into a standardized boundary delimiter, conversion of maqaf-linked forms into token-separated units, and preservation of word and phrase boundaries as explicit structural markers.

The goal of preprocessing is to produce a stable and reproducible phonetic stream while preserving structural information necessary for boundary-sensitive analysis.

2.3 Phonetic representation and transliteration

A deterministic rule-based phonetic transliteration is used to convert Hebrew orthography into a standardized sequence of phonetic units. This transformation serves three purposes, namely reduction of orthographic variation into a representation suitable for quantitative analysis, integration of consonantal and vocalic information into a single analytic space, and transparency and inspectability independent of the original script.

The representation is not intended as a full phonological reconstruction, but as a controlled analytic encoding that preserves distinctions relevant to recurrence, clustering, and correspondence structure.

Vowels are mapped into a five-class system: {a, e, i, o, u}. This includes consistent treatment of vowel letters and common vocalic realizations, with the goal of maintaining a stable vowel layer for comparative analysis rather than encoding full phonetic detail.

Consonants are represented using a standardized mapping preserving key phonemic contrasts (e.g., b/v, k/kh, p/f, sh/s, ts, q). The letter *yod* is treated as a consonant. Multi-character units (sh, kh, ts) are treated as atomic units in all analyses.

Certain weak consonants are omitted when they do not contribute a stable phonetic realization, while consonants affecting phonetic structure are retained. In particular, final ḥ (e.g., *ruaḥ*) is represented such that the vowel precedes the consonant.

Sheva is resolved deterministically: word-initial sheva is mapped to *e*, while non-initial sheva is treated as non-vocal unless required for syllabic support.

The resulting phonetic representation forms a continuous stream in which spaces are used to mark word boundaries, while a dedicated delimiter "|" marks phrase-final boundaries and is attached to the final word of each phrase.

Maqaf (־) is removed and replaced by whitespace. Cantillation marks and punctuation are removed, except for the phrase-boundary delimiter.

This unified phonetic stream serves as input for all analyses. All transliteration rules are deterministic and applied uniformly across corpora.

2.4 Rhyme unit definition (Layer C)

Rhyme units are defined directly from the phonetic stream based on vowel positions. For closed syllables, each vowel defines a unit extending from the vowel to all subsequent consonants up to, but not including, the next vowel. For open word-final syllables, the unit is defined as a consonant–vowel sequence (preceding consonant plus vowel), even if the consonant participates in a previous unit. Units of length less than two are excluded. This definition is used consistently across all Layer C analyses.

2.5 Multi-layer analytical framework

Phonetic organization is modeled as a multi-layer system composed of three interacting levels:

Layer A — Consonantal structure

At the consonantal level, local organization is captured through recurrence and overlap of consonantal units and short sequences. Consonants are treated as atomic units (including sh, kh, ts), and short sequences ($k = 2$–4) are analyzed as contiguous units extracted from the phonetic stream.

The analysis considers recurrence of short phonetic sequences ($k = 2$–4), clustering of repeated sequences, and the formation and stability of locally dominant consonantal sets.

Three primary measures are used: recurrence density, defined as the proportion of sequences that occur at least twice within the analysis window or stream; gap, defined as the distance (in phonetic units) between successive occurrences of identical sequences; and overlap, defined as the stability of dominant consonantal configurations across adjacent segments. Distances are computed over the continuous phonetic stream (Section 2.3). Statistical significance is assessed relative to block-randomized null models (Section 2.7), which preserve local phonetic composition while disrupting larger-scale organization.

This layer evaluates whether consonantal structure exhibits clustering and persistence beyond what is expected from local phonetic distributions.

Layer B — Phonetic chain structure

This layer captures the organization of phonetic sequences across the stream, focusing on how recurrence patterns extend beyond local contexts and interact with structural segmentation.

The analysis considers local k-sequence recurrence under sliding conditions, participation of sequences in recurring chains, and sensitivity of recurrence structure to segmentation boundaries.

Gap is defined as the distance (in phonetic units) between successive occurrences of identical k-sequences. Recurrence is evaluated over the continuous phonetic stream (Section 2.3), with optional restriction to segments defined by phrase boundaries.

Baseline measurements are compared to both full randomization (global shuffle) and segment-level randomization preserving boundaries.

Statistical evaluation follows the same framework as in Layer A, using block-randomized null models (Section 2.7) and permutation-based Z-scores.

This layer tests whether phonetic dependencies are organized in relation to structural segmentation, rather than being reducible to local phonetic constraints alone.

Layer C — Syllabic and rhyme structure

Syllabic and rhyme structure is defined over the phonetic stream described in Section 2.3. Vowels are represented as {a, e, i, o, u}, and consonants as atomic units, with digraphs (sh, kh, ts) treated as single phonetic units. The letter y is treated as a consonant.

Rhyme units are defined deterministically from vowel positions. For closed syllables, each vowel defines a unit extending from the vowel to all subsequent consonants up to, but not including, the next vowel (V + C*). For open word-final syllables, the unit is defined as a consonant–vowel (CV) sequence using the preceding consonant and final vowel. Units of length less than two (after extraction) are excluded. These units define the correspondence structures analyzed as rhyme-like correspondences.

All analyses operate on these units in two modes of application: stream-based analyses (syllable_flow, syllable_distance, window_density), in which units are extracted over the continuous phonetic stream while ignoring word and phrase boundaries; and boundary-based analyses (word_final, phrase_final), in which units are evaluated relative to word and phrase boundaries defined in the phonetic stream.

Distances between recurrences are computed over the sequence of units with a maximum lag parameter L (typically L = 40). Null models are constructed using block-shuffled sequences (block size = 50 unless otherwise noted), preserving local phonotactic structure while disrupting long-range organization. Statistical significance is assessed via permutation tests (typically 200 permutations, seed = 1).

Window-based analyses use sliding windows (e.g., W = 1000, step = 500) to evaluate the continuity of correspondence activity across the text. This part of the framework tests whether syllabic and rhyme-like correspondences exhibit structured recurrence and large-scale organization beyond local phonetic expectations.

2.6 Statistical definitions

The following general measures are used across layers: recurrence density, defined as the proportion of eligible units or sequences participating in recurrence; gap, defined as the number of phonetic units between repeated occurrences; and correspondence rate, defined as the proportion of positions exhibiting a match under the specified rule set.

All measures are computed over the phonetic stream or its derived representations, depending on the layer.

Statistical deviation from null expectations is quantified using $Z = (\text{observed} - \text{mean\_null}) / \text{sd\_null}$, where the null distribution is obtained from repeated randomized realizations of the phonetic stream under controlled conditions (Section 2.7).

Z-scores quantify deviation from null expectations and are not interpreted as direct measures of effect magnitude across datasets or analytical layers.

2.7 Null models and controls

To test whether observed structure exceeds local phonetic expectations, null models are constructed through controlled randomization.

The primary null model uses block-wise shuffling of the phonetic stream. The stream is partitioned into contiguous blocks (measured in phonetic units), which are randomly permuted while preserving internal order within each block. This preserves local phonetic composition while disrupting long-range dependencies.

Unless otherwise noted, block size is set to 50 phonetic units, with additional block sizes tested for robustness (Section 2.8).

For segmentation analyses, additional controls are applied, namely full randomization (global shuffle of the phonetic stream) and segment-level randomization, in which segments defined by phrase boundaries are permuted while preserving internal structure. These controls distinguish effects driven by structural organization from those arising purely from local phonetic distributions.

Null distributions are generated via permutation (typically ≥200 realizations, seed = 1 unless otherwise specified). Statistical significance is assessed using Z-scores computed relative to the null distribution.

2.8 Robustness analysis

Key analyses are repeated under parameter variation, including sequence length ($k = 2$–4), block size in null models, and window size and distance parameters in syllabic analyses (Layer C). An effect is considered robust if its direction, magnitude, and qualitative structure persist across these variations. These robustness checks include both baseline and stress-test parameter settings, as documented in the repository protocols.

2.9 Comparative framework

The analytic framework is applied identically to the Pentateuch and the modern Hebrew corpus. The comparison evaluates differences in the magnitude of effects, consistency across scales, sensitivity to segmentation, and degree of large-scale coordination. The comparison is interpreted as graded rather than binary and serves as a reference for distinguishing structured organization from general properties of Hebrew phonetic sequences.

2.10 Reproducibility and implementation

All preprocessing rules, phonetic encoding procedures, analysis scripts, and parameter settings are implemented computationally; repository details are provided in the Data and Code Availability section.

The definitions provided in Sections 2.3–2.5 fully specify the phonetic representation and analytical units used in all analyses.

2.11 Scope and limitations

The analysis focuses on the phonetic stream derived from a standardized encoding and does not incorporate prosodic timing, cantillation performance, stress modeling, or community-specific pronunciation variation. The transliteration is a controlled analytic representation rather than a full phonological reconstruction. The aim is to test whether large-scale phonetic organization is detectable under a conservative and reproducible encoding.

3. Results

3.1 Overview

Across all analytical layers, the Pentateuch exhibits phonetic patterning that significantly deviates from randomized controls preserving local phonetic distributions. These effects are observed across multiple layers and scales, indicating structured organization in the phonetic stream.

Rather than appearing as isolated phenomena, the observed patterns form a coherent structure across consonantal, chain-based, and syllabic levels, suggesting coordinated organization within a multi-layer phonetic system.

3.2 Layer A — Consonantal structure

Analysis of consonantal sequences ($k = 2–4$) reveals elevated recurrence density, substantially reduced inter-occurrence distances, and increased overlap of dominant consonantal configurations relative to block-randomized controls.

The effect is observed consistently across all books of the Pentateuch and is substantially weaker in the comparison corpus.

The primary distinguishing feature is not the amount of repetition, but its spatial organization: repeated phonetic sequences cluster significantly more tightly than expected under local-preserving randomization.

These effects are robust across parameter variation, including sequence length (k), block size, and window configuration, with consistently large positive Z-scores across settings.

Figure 1 illustrates the magnitude of the consonantal recurrence effect across the Pentateuch and the comparison corpus.

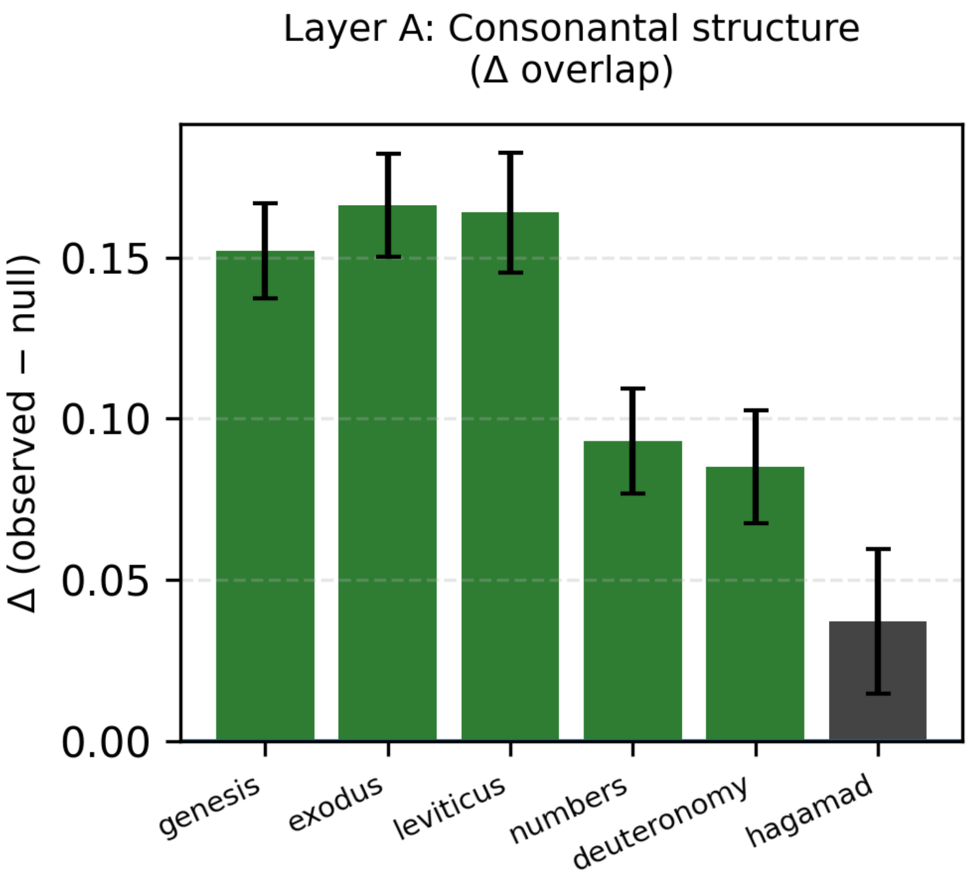


Figure 1. Consonantal recurrence structure (Layer A).
Bars show deviation from the null model (observed − mean null) in overlap of locally dominant consonantal configurations for each book of the Pentateuch and the modern Hebrew corpus. Error bars indicate standard deviation under block-randomized null models. All Pentateuch books exhibit substantially higher overlap than the control corpus, indicating stable clustering of consonantal patterns beyond local phonetic expectations.

In addition to sequence-level recurrence, consonantal configurations form locally stable dominant sets across segments. These sets exhibit elevated mass and adjacent stability, indicating recurring phonetic regimes rather than isolated repetitions.

These results indicate that the consonantal layer forms a stable local pattern of clustering.

3.3 Layer B — Boundary-sensitive chain structure

Boundary-based analysis shows that phonetic clustering is strongly dependent on textual segmentation.

Gap measures differ substantially between baseline and boundary-disrupted conditions across all tested parameters. For example, for k = 2, baseline gaps (~400) decrease to ~10 under boundary conditions, and for k = 3, baseline gaps (~2000) decrease to ~12.

Figure 2 illustrates the magnitude of the boundary effect across the Pentateuch and the comparison corpus under segment-level controls.

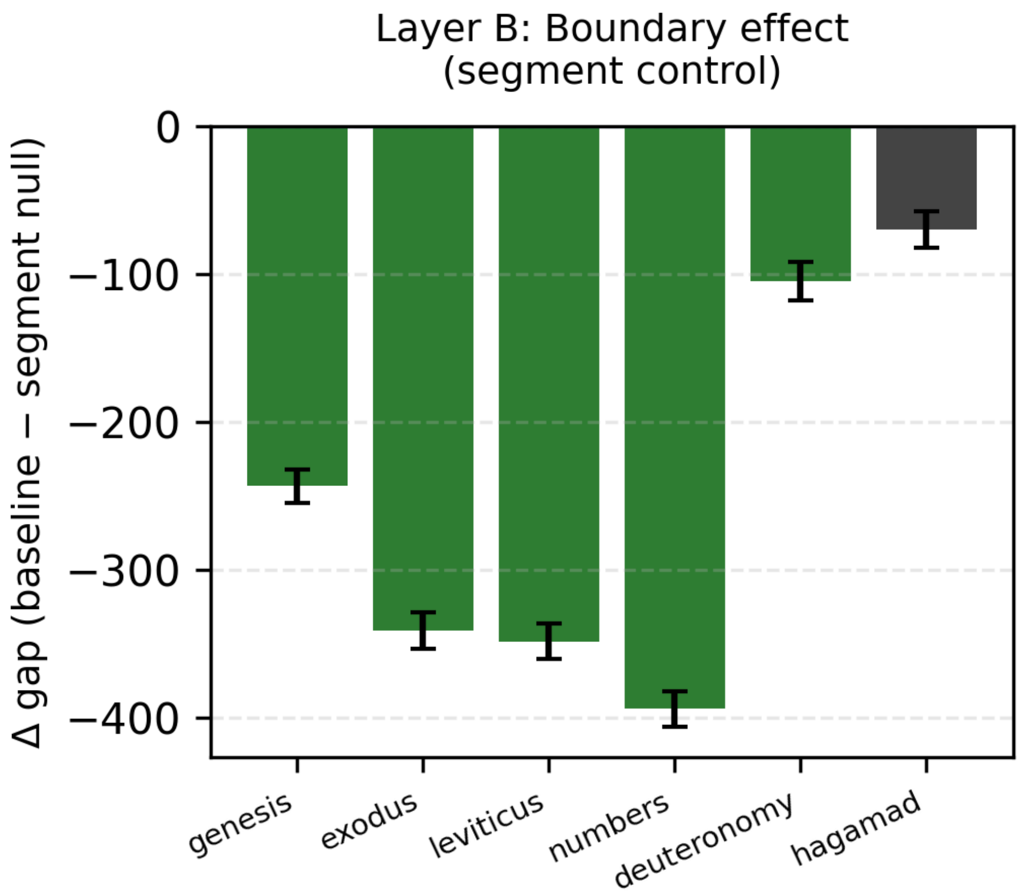


Figure 2. Boundary-sensitive phonetic structure (Layer B).
Bars show the difference in gap (baseline − segment-level null) for each book of the Pentateuch and the modern Hebrew corpus. Error bars indicate standard deviation under segment-level randomization. Lower gap values correspond to tighter clustering of repeated sequences. The Pentateuch exhibits substantially larger deviations from the null model, indicating strong alignment of phonetic clustering with structural segmentation, whereas the control corpus shows weaker and less consistent effects.

This pattern holds across k = 2–4, indicating that the effect is not an artifact of sequence length.

Comparison with modern Hebrew prose further shows that clustering is consistently tighter in the Pentateuch (e.g., boundary gap ≈10–15 in the Pentateuch vs ≈13–22 in modern text).

Z-scores under full randomization indicate strong deviation from null expectations ($Z \approx -8$ to -32), corresponding to substantial reduction of structured recurrence under shuffle. Segment-level controls also show strong effects (Pentateuch Z values reaching −30, versus approximately −5 to −6 in the modern corpus), indicating sensitivity to segment ordering.

These results demonstrate that phonetic clustering is not only present but systematically aligned with structural segmentation, indicating that phonetic dependencies are coordinated with higher-level textual organization.

Additional analysis shows that anchor-based and sequential formulations produce highly similar patterns, indicating that the observed structure does not depend on a specific representation of local context.

Local chain recurrence, anchor-based recurrence, and boundary-sensitive controls converge on the same conclusion: phonetic dependencies are both locally clustered and structurally aligned.

3.4 Layer C — Syllabic and rhyme structure

3.4.1 Flow and propagation

Syllabic correspondence analysis reveals strong and scale-dependent propagation of rhyme-tail recurrence across the phonetic stream.

At the baseline scale ($L = 20$), all five books exhibit strongly elevated Z-scores (Genesis $Z = 17.7$, Exodus $Z = 21.7$, Numbers $Z = 18.8$, Leviticus $Z = 15.3$, Deuteronomy $Z = 9.6$), while the modern control corpus shows only a weak effect ($Z \approx 2$).

The effect varies systematically with analysis scale: at $L = 10$, Z-scores range approximately from 5 to 14; at $L = 20$, from 9 to 22; and at $L = 40$, from 15 to 30. This systematic increase with analysis scale indicates that recurrence remains detectable across larger distances and that the separation from null expectations becomes more pronounced at larger correspondence distances.

The effect remains stable across null configurations (block sizes 50 and 80), with high Z-scores in the Pentateuch and substantially weaker effects in the control corpus.

These results indicate that rhyme-tail recurrence is a large-scale property of the phonetic stream.

In addition to statistical significance, absolute correspondence measures (e.g., density and match rates) are consistently higher in the Pentateuch than in the control corpus.

Figure 3 illustrates the magnitude of syllabic correspondence across the Pentateuch and the comparison corpus.

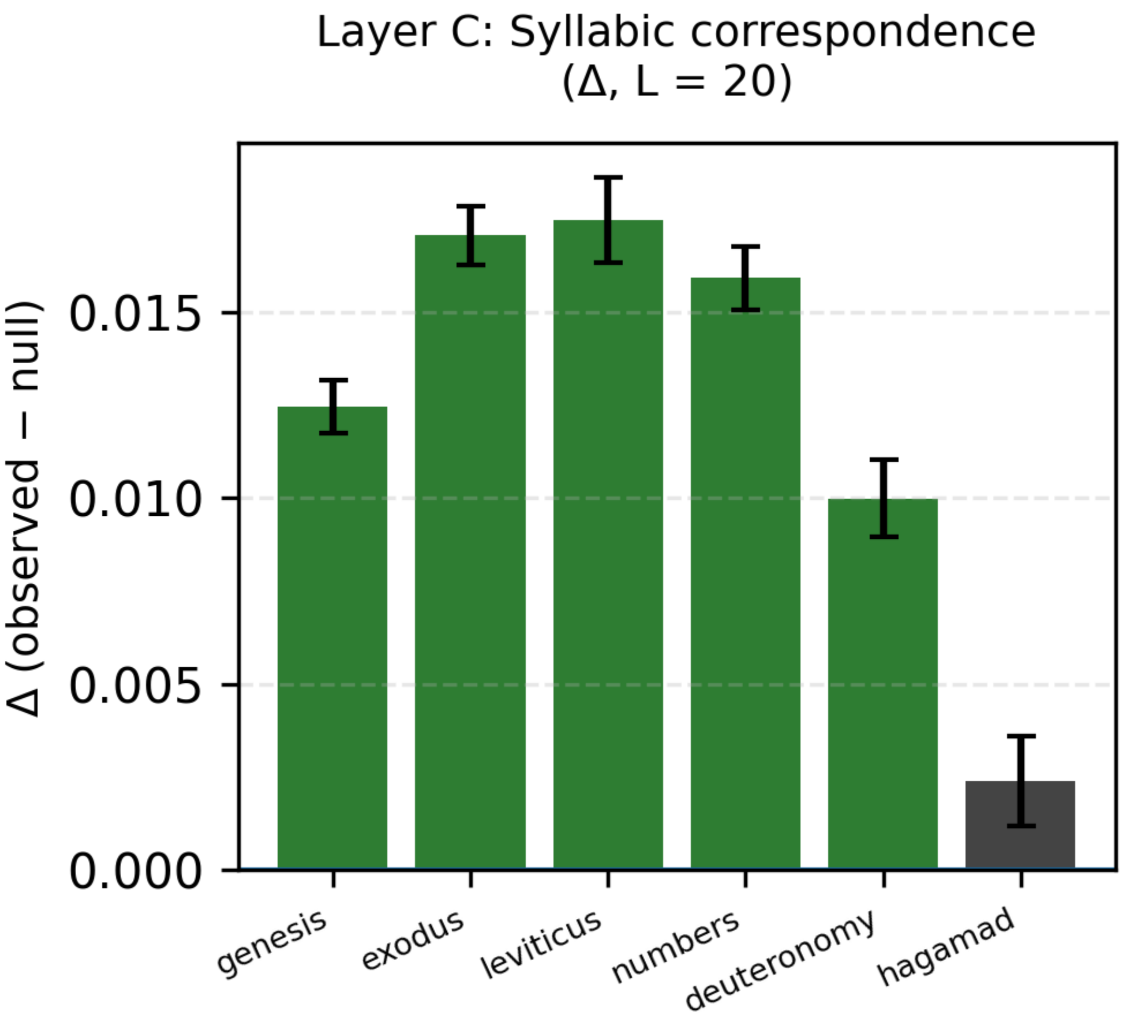


Figure 3. Syllabic correspondence structure (Layer C).
Bars show deviation from the null model (observed − mean null) for syllabic correspondence (rhyme-like recurrence) at L = 20 across the Pentateuch and the modern Hebrew corpus. Error bars indicate standard deviation under block-randomized null models. All Pentateuch books exhibit strong positive deviations, while the control corpus remains close to the null expectation, indicating substantially weaker large-scale correspondence structure.

The comparatively weaker signal observed in Deuteronomy may reflect differences in rhetorical structure, such as extended speech-like passages, though this requires further investigation.

3.4.2 Distance structure

Rhyme recurrence distances reveal a strongly non-uniform and structured distribution in the Pentateuch.

Across all five books, global Z-scores are strongly elevated (e.g., Genesis Z=17.7, Exodus Z=21.7, Numbers Z=18.8), while the modern control corpus shows only weak effects (Z≈2).

Crucially, recurrence is not evenly distributed across distances. Instead, it concentrates at specific preferred intervals. Prominent peaks consistently appear in the ranges ~20–30 and ~35–40 units, with additional secondary peaks at shorter distances.

Individual distance peaks reach substantial positive deviations from null expectations (e.g., Z ≈ 8 at specific distances), indicating highly structured and repeatable recurrence intervals.

These peaks persist across parameter variation, indicating that they reflect intrinsic structural organization rather than analysis artifacts.

This pattern demonstrates that rhyme recurrence operates at structured, repeatable distances, rather than arising from random local clustering.

3.4.3 Word-final concentration

Phonetic correspondences show strong concentration at word-final positions.

Observed Z-scores range approximately from 17 to 31 across books, with modern Hebrew showing substantially lower values (~6–7).

The effect remains strong under stricter null models (e.g., Z ≈ 14.6 under larger block sizes), indicating robustness.

This demonstrates that phonetic correspondences are preferentially localized at word boundaries.

3.4.4 Phrase-final network

Phrase-level analysis reveals a statistically significant network of correspondences linking phrase endings.

Z-scores range approximately from 8 to 16 across books, while modern Hebrew shows values close to zero (~1).

This indicates that phrase-final units are systematically connected through phonetic correspondence in the Pentateuch, while such structure is largely absent in the comparison corpus.

3.4.5 Continuous distribution (Window Density)

Window-based analysis indicates that rhyme activity is continuously maintained across the Pentateuch.

Mean rhyme density per window is consistently higher than in the control corpus (Pentateuch ≈ 0.021–0.023; modern ≈ 0.016), with broader variance, indicating structured modulation across the text.

The difference corresponds to a substantial relative increase in local rhyme density compared to the control corpus.

In contrast, the modern corpus exhibits lower density and reduced variance, with extended low-density regions that are not comparably evident in the Pentateuch.

This demonstrates that rhyme-like correspondence is not confined to isolated passages, but is distributed broadly across the text.

This pattern remains stable across window and radius settings, indicating that the result is not driven by a specific parameter choice.

The pattern is illustrated in Figure 4, which shows the spatial distribution of local rhyme density across the Pentateuch.

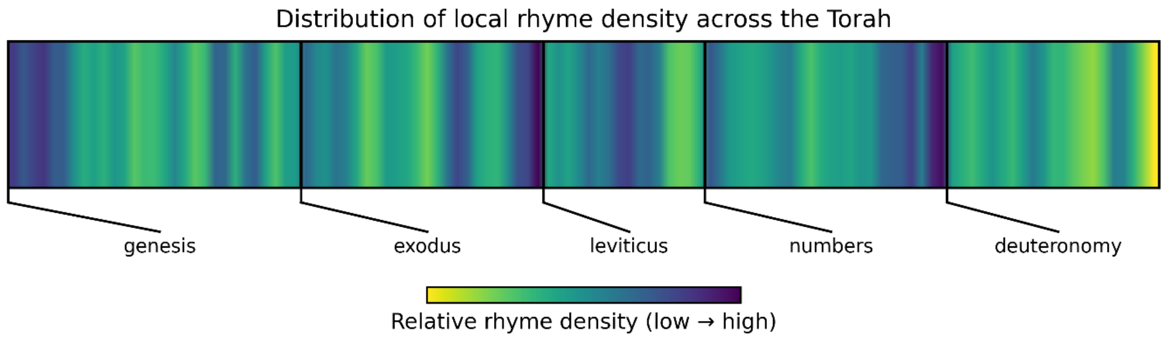


Figure 4. Local rhyme density across the Pentateuch.
The figure shows normalized rhyme density computed over sliding windows across the continuous phonetic stream. Color intensity reflects relative local density. The distribution illustrates that rhyme-like correspondence is broadly maintained across the text, with relatively few extended low-density regions.

3.5 Cross-layer comparison with modern Hebrew prose

Across all layers, modern Hebrew literary prose exhibits weaker phonetic patterning, reduced clustering, and limited large-scale coordination.

While similar tendencies are detectable, they are smaller in magnitude, less consistent across scales, and lack the coordinated interaction between layers observed in the Pentateuch.

This comparison supports a graded interpretation in which related phonetic tendencies may exist in the language, but are not organized into a coherent multi-layer structure in modern prose.

3.6 Summary of results

Across all layers, the Pentateuch exhibits phonetic organization that significantly exceeds expectations under local-preserving randomized controls, remains robust across parameter variation (k, block size, window scale), is strongly sensitive to structural segmentation and segment ordering, shows non-uniform distributions with positional concentration, and exhibits structured recurrence distances together with broad distribution across the text. Taken together, these results indicate that phonetic organization in the Pentateuch is most consistently interpreted as a coordinated multi-layer system operating at a global scale.

4. Discussion

4.1 Principal findings

The present study provides evidence for a coherent multi-layer system of phonetic organization operating across the Biblical Pentateuch. Across all analytical layers, phonetic patterning in the text is clearly structured and shows strong alignment with segmentation.

At the consonantal level, recurrence exhibits structured clustering beyond local expectations. At the chain level, phonetic dependencies are organized in relation to segmentation boundaries. At the syllabic level, correspondences display non-uniform distributions, structured recurrence distances, and systematic positional concentration.

These effects are consistent across parameter variation and persist under randomized controls that preserve local phonetic composition. Taken together, they indicate that phonetic organization operates as a coordinated system spanning multiple structural levels, rather than as a collection of independent phenomena.

4.2 Relation to linguistic structure

A central question is whether the observed patterns can be explained solely by properties of the Hebrew language, such as morphology or phonological constraints. While such factors may contribute to local phonetic regularities, several aspects of the results argue against a purely local explanation. First, the observed effects consistently exceed expectations under randomization procedures that preserve local phonetic distributions. Second, the strong sensitivity to structural segmentation indicates that phonetic dependencies are aligned with higher-level textual organization. Third, their consistency across the corpus suggests organization beyond isolated local phenomena. Taken together, these findings indicate that, although local linguistic factors may contribute to baseline recurrence, they do not appear sufficient to account for the observed large-scale coordination and segmentation alignment. The observed phonetic organization is therefore unlikely to be fully reducible to morphology or local phonological constraints alone but reflects additional structure operating at the level of the text.

4.3 Comparison with modern Hebrew prose

Comparative analysis shows that modern Hebrew literary prose exhibits related phonetic tendencies, but in a weaker and less consistently organized form, with only occasional local coordination.

This suggests that the mechanisms enabling phonetic recurrence may exist more generally in the language, but that the Pentateuch differs in the degree of large-scale coordination and integration of these mechanisms across structural levels.

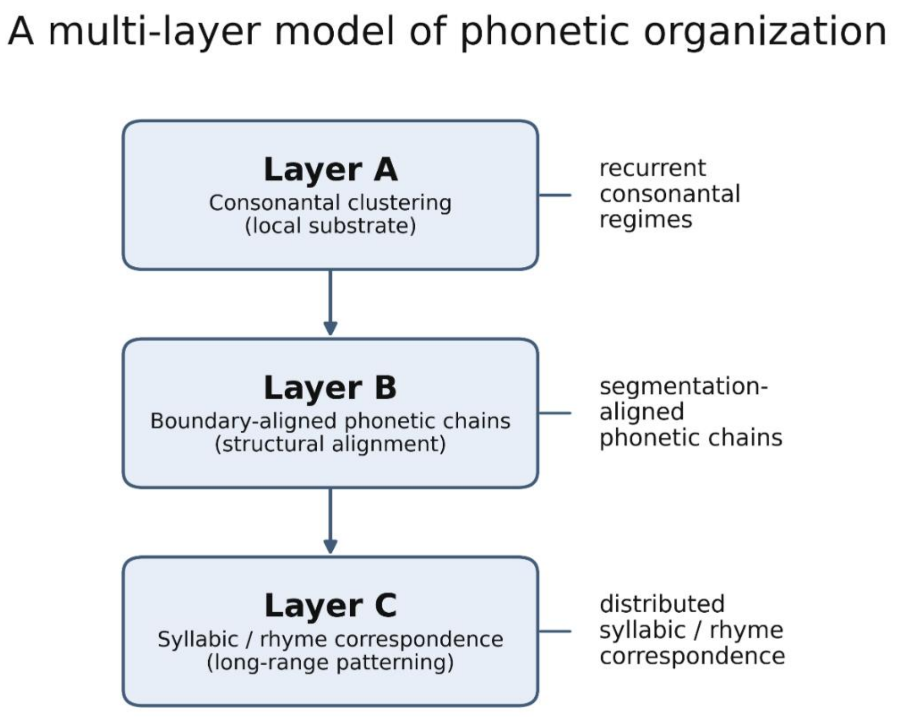
The comparison therefore supports a graded interpretation, in which similar tendencies may be present across texts, but only in the Pentateuch do they form a coherent multi-layer system. The difference is thus primarily quantitative and organizational rather than strictly categorical.

The comparison with modern prose rather than poetic corpora is intentional, as the Pentateuch is conventionally classified as narrative text. Demonstrating structured phonetic organization exceeding that of literary prose is therefore central to the present argument.

The comparison is limited to a single modern corpus and should be interpreted as indicative rather than exhaustive.

4.4 Toward a model of phonetic organization

The combined results across layers suggest that phonetic structure in the text is not composed of isolated effects, but arises from the interaction of multiple levels of organization. This interaction is summarized schematically in Figure 5.



Interaction across layers yields structured large-scale phonetic organization.

Figure 5. A multi-layer model of phonetic organization.
The three analytical layers are interpreted as interacting levels of phonetic structure. Their interaction defines a constrained phonetic environment in which large-scale structured patterning emerges.

In this framework, Layer A provides a local consonantal substrate through clustered recurrence. Layer B aligns phonetic dependencies with structural segmentation. Layer C expresses distributed syllabic and rhyme-like correspondences across the text. As a result, phonetic units are not independently distributed, but organized in a way that increases the likelihood of repeated and positionally structured correspondences.

Under this interpretation, large-scale phonetic patterning can be understood as an emergent property of a multi-layer system rather than as the result of isolated or explicitly encoded correspondences. The system does not require direct specification of individual matches; instead, it creates conditions under which such correspondences arise across multiple scales.

4.5 Relation to poetic structure

The observed organization shares functional properties with known forms of poetic structuring, including recurrence, positional concentration, and non-uniform distribution of sound correspondences [1–3]. However, unlike traditional poetic forms, which are typically defined at the level of lines or stanzas, the patterns observed here extend across a continuous textual stream.

Taken together, the distance and window analyses reveal two complementary properties of the system. First, rhyme recurrence is concentrated at specific, repeatable distances, indicating structured long-range organization. Second, rhyme activity is continuously maintained across the text, with highly continuous distribution and relatively few extended low-density regions.

This combination implies that the system is neither purely local nor randomly distributed. Instead, it reflects a globally organized phonetic structure operating across multiple scales, consistent with a systematically organized compositional framework.

These findings suggest that phonetic organization in extended texts may operate as a distributed system rather than as a set of discrete formal units. The results are therefore consistent with an extension of conventional models of poetic structure, in which sound patterning can occur across multiple scales simultaneously and without explicit line-based segmentation [4, 5].

This type of distributed rhyme organization is more similar to multi-scale rhyme systems observed in certain modern oral-poetic forms than to traditional line-based poetic structures.

4.6 Why such structure may be difficult to detect

One implication of the present findings is that large-scale phonetic organization may remain difficult to detect under standard analytic assumptions.

First, if similar phonetic tendencies occur at a low level in ordinary language, they may be interpreted as baseline linguistic properties rather than as components of a structured system [9]. Second, if phonetic patterning is distributed rather than localized, it may not be readily captured by methods that focus on short textual segments. Third, the absence of explicit lineation or conventional poetic markers may obscure the presence of structured sound correspondences [1, 2].

These factors together may contribute to the under-recognition of multi-scale phonetic organization in extended texts.

4.7 Limitations

Several limitations should be noted. The phonetic representation used in this study is a simplified and standardized encoding designed for reproducible analysis rather than full phonological reconstruction.

In addition, the analysis does not incorporate prosodic timing, stress patterns, or performance-based features, which may further influence phonetic realization in recited contexts.

The study also focuses on a single primary corpus. While comparisons are made with modern Hebrew prose, broader cross-corpus and cross-linguistic generalization remains to be investigated.

4.8 Future directions

Future work may extend the present framework in several directions, including applying the same analytic approach to additional corpora, such as other parts of the Hebrew Bible and texts in other languages; incorporating prosodic and performance-based features into the analysis; refining phonetic representations to capture additional dimensions of variation; and developing formal models of the interaction between phonetic layers and their contribution to large-scale organization. Such extensions may help determine whether the type of phonetic organization described here represents a broader class of structured textual systems. The present results establish the existence of structured phonetic organization; further work is required to determine its origin and generative principles.

Data and Code Availability

All data processing procedures, phonetic encoding rules, and analysis scripts used in this study are implemented computationally and are available in a public version-controlled repository: https://github.com/HaShira-Lab/torah-phonetic-architecture

The primary corpus (Biblical Pentateuch) was obtained programmatically via the Sefaria API (version: Tanach with Ta'amei HaMikra), and the comparison corpus (**HaGamad Baal HaChotem**) was obtained from the Ben Yehuda Project. As redistribution of source texts may be restricted, the repository provides scripts and metadata necessary to reconstruct the dataset from the original sources.

Detailed protocols for preprocessing, phonetic encoding, analytical procedures, and statistical evaluation are provided in the `/protocols/` directory. Scripts for reproducing all reported figures are included.

All reported results are reproducible using the provided code and parameters. Randomized procedures are performed with fixed seeds and documented configurations to ensure full reproducibility.

References

1. Jakobson R. Linguistics and Poetics. In: Sebeok TA, editor. Style in Language. Cambridge (MA): MIT Press; 1960. pp. 350–377.

2. Attridge D. Poetic Rhythm: An Introduction. Cambridge: Cambridge University Press; 1995.

3. Fabb N. Linguistics and Literature. Oxford: Blackwell; 1997.

4. Fabb N. Language and Literary Structure: The Linguistic Analysis of Form in Verse and Narrative. Cambridge: Cambridge University Press; 2002.

5. Manning CD, Schütze H. Foundations of Statistical Natural Language Processing. Cambridge (MA): MIT Press; 1999.

6. Alter R. The Art of Biblical Narrative. New York: Basic Books; 1981.

7. Kugel JL. The Idea of Biblical Poetry: Parallelism and Its History. New Haven: Yale University Press; 1981.

8. Berlin A. The Dynamics of Biblical Parallelism. Bloomington: Indiana University Press; 1985.

9. Ellis NC. Frequency Effects in Language Processing: A Review with Implications for Theories of Implicit and Explicit Language Acquisition. Stud Second Lang Acquis. 2002;24(2):143–188.

10. Glikshtern E. "…and now write down the Song…" – The Intrinsic Musical Content of the Holy Scriptures and a New Method for Deciphering the Taʿamei ha-mikra. In: Nemtsov J, Klein JM, editors. Magic Violin: Jüdische Musikkultur und Moderne. Wiesbaden: Harrassowitz Verlag; 2024. p. 133–194.

11. Moretti F. Distant Reading. London: Verso; 2013.